Cambridge
O Level

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Cambridge Ordinary Level

ENVIRONMENTAL MANAGEMENT **5014/21**

Paper 2 **May/June 2017**

MARK SCHEME

Maximum Mark: 60

Published

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Question	Answer	Marks
1(a)(i)	*any two of:* damage / interfere, with farmland / eq; so less food available / eq; *reference to* risk of pollution of Nile; risk of flooding homes; difficult to travel into the city; easier / more space, to the east; AVP, e.g. unsuitable ground for building;	**2**
1(a)(ii)	estimate between 100–200 (km^2);; (*if answer incorrect, allow one mark for* use of scale *[1]*);	**2**
1(a)(iii)	*any two of:* only desert; not used for farming; unused land; more space available for building; connects to existing housing / old city; connects to services / named services; low transport costs; no risk of flooding; AVP;	**2**
1(a)(iv)	*any three of:* roads / transport; electricity; telecoms; water supply; sewage removal; hospitals / healthcare / clinics; schools / education; *reference to* street lights / waste collections / other example; law enforcement / other named emergency service;	**3**

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Question	Answer	Marks
1(a)(v)	*any two of:* cooling effect; *reference to* transpiration; shade; windbreak; absorbs some air pollutants; scenic value / pleasant environment / eq;	**2**
1(b)	*any three of:* for irrigation; so crops can grow all year; cheap / easy access to, water supply; easy to keep, animals / livestock; close to, population / farm workers; transport, crops / goods, to people; rich / fertile soil;	**3**
1(c)(i)	*any two of:* bacteria fix nitrogen gas from atmosphere / eq; pea plants do not need fertiliser; *reference to* amino acids / proteins / DNA / enzyme;	**2**
1(c)(ii)	5 / 4.6 / 4.58 / 4.583; g;	**2**
1(c)(iii)	5 OR 3–8;	**1**
1(c)(iv)	2400;; (*if answer incorrect, allow one mark for correct method, e.g.* 60 × 40 *[1]*)	**2**
1(d)(i)	*next two pairs circled* 4+6 AND 9+5; correctly drawn on grid;;	**3**

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Question	Answer	Marks
1(d)(ii)	more plants / samples (3 not 2); more areas of field (10 areas rather than 1); random sampling; many samples remove bias / eq; anomalies / outliers, can be identified; and removed from average or have less effect; *reference to* validity;	**3**
1(e)(i)	*4 correct [2]* *2 to 3 correct [1]* 5, 2, 3, 1;;	**2**
1(e)(ii)	**R** and *any two of:* **highest** values for, BOD; bacteria; chromium; iron; allow a reason, e.g. not enough oxygen means organisms will die / chromium is toxic / etc.;	**2**
1(e)(iii)	**T** and low / no, BOD / bacteria count / no metal content;	**1**
1(e)(iv)	**R** and highest concentration of, chromium / iron;	**1**
1(e)(v)	*any two of:* organic matter / sewage / fertiliser, present; more bacteria / bacteria feed on this / multiply / eq; bacterial respiration increases / respire more oxygen; so oxygen used up / eq;	**2**
1(f)(i)	129 / 128.6 / 128.57 / 128.571(%);; (*if answer incorrect, allow one mark for* 4.5 e.g. 8.0–3.5 = 4.5, 4.5 / 3.5 × 100 *[1]*)	**2**
1(f)(ii)	to check there were no changes in temperature / make sure it was not an important factor in this canal;	**1**

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Question	Answer	Marks
1(f)(iii)	*any two of:* dissolved oxygen decreased at site 3; dropped by a half / by 4.4 (between site 2–3); organic matter / sewage / fertiliser / other valid effluent, e.g. tannery waste; so bacteria increase in numbers and use up oxygen / BOD increase;	**2**
1(g)	*allow any three valid suggestions, such as:* more monitoring of water sources; strict laws about water quality; enforcing the laws; take polluters to court; polluter pays for clean up; AVP;	**3**
2(a)(i)	add all the responses for each question; express as a % of total number of questionnaire or out of 100;	**2**
2(a)(ii)	so decision makers can compare results (of different living areas); to find out if only people near the canal were in favour / ORA;	**2**
2(a)(iii)	*any two of:* reason why (government) need the money / eq; government can find out if people, approved of the project / value the project; give people / nation, a sense of ownership of project / eq; AVP;	**2**
2(a)(iv)	*any two of:* close to, canal / new roads, for easy transport; low cost of transport; encourages, import / export / trade; creates new jobs / canal already has skilled workers, so readily available / eq; provides source of water for industries;	**2**
2(a)(v)	tax breaks / investment grants / interest free loans / subsidies / eq or AVP;	**1**

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Question	Answer	Marks
2(a)(vi)	*any three of:* *for*: saves, money / time; so more ships will use canal; more canal fees earned; AVP; *against:* canals already save days / weeks anyway (not going around Africa); new route only saves 8 hours; may not be demand for these improvements; AVP;	**3**
2(b)(i)	*any two of:* swim; flow / drift, in current; in ships bilges; cargo; stuck on hull; smuggled / tourists; AVP;	**2**
2(b)(ii)	*any three of:* disrupt food chain / web; *reference to* competition; e.g. effect of invasive producer or consumer; lack of natural predators; so population explosion; or population crash; habitat destruction; decrease in biodiversity; credit valid aquatic example; can carry, disease / pathogen / toxin;	**3**

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